PROBLEM 15.19 **Working drawing (in.)**

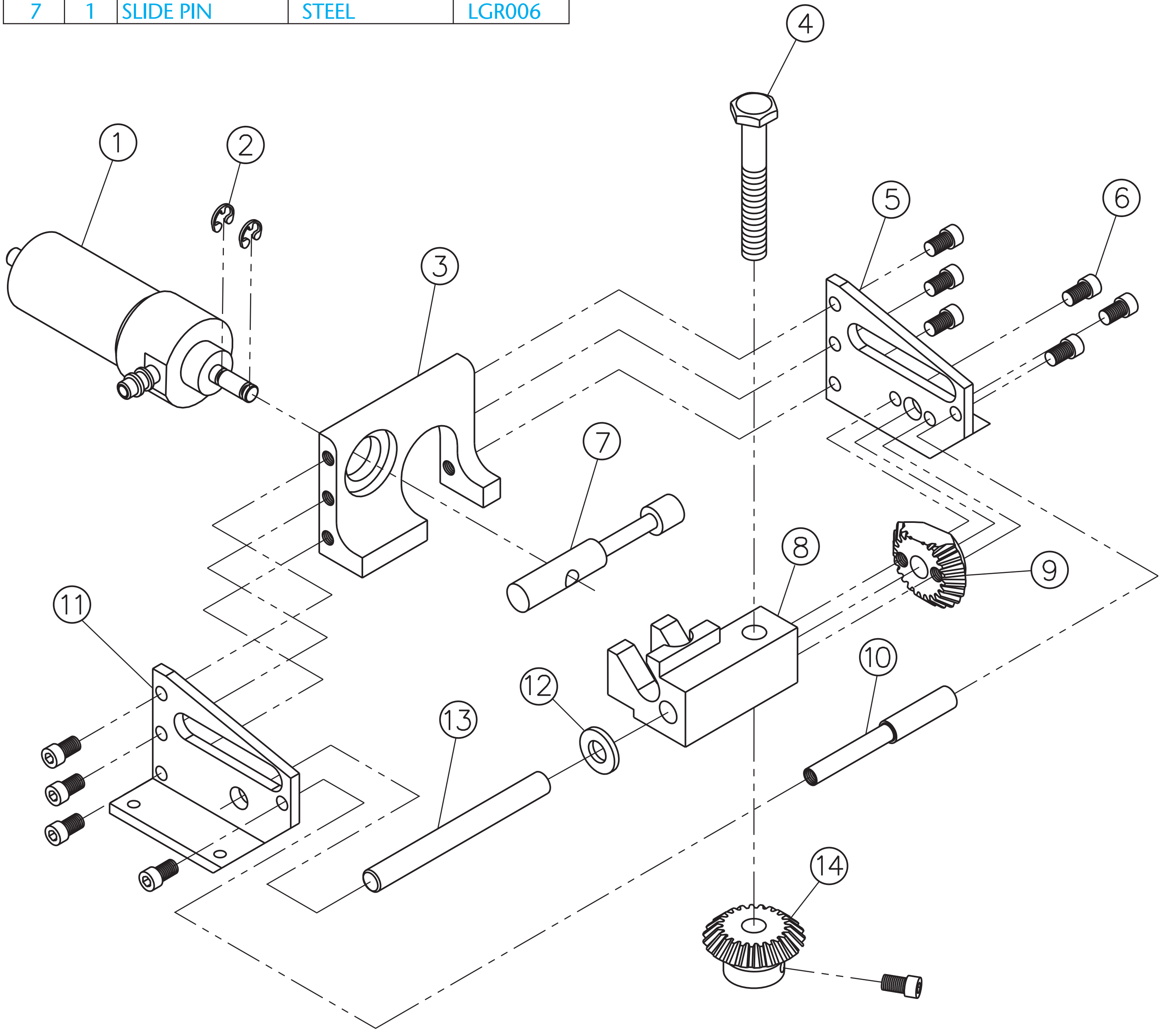
Assembly Name: Landing Gear Retract Assembly

SPECIFIC INSTRUCTIONS: This problem is challenging and contains applications that include gears. You should study gears and related drafting practices covered in Chapter 16 before completing this problem.

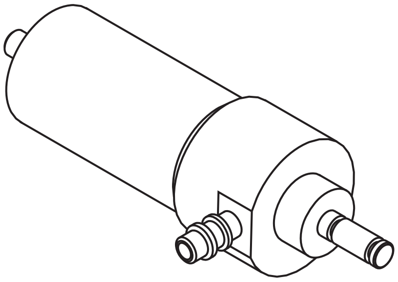
Prepare a complete set of working drawings with one detail drawing per sheet and the assembly and parts list on another sheet. When preparing the assembly drawing, use separate balloons for each part in each view or use only one balloon in the view that most clearly identifies the part. This is an engineering prototype project. Errors are likely in this problem. Verify dimensions during assembly. Establish tolerances between mating parts. There are several purchase parts in this assembly. Manufacturer research is needed to locate available products.

PARTS LIST

ITEM	QTY	NAME	DESCRIPTION	PART NO.
1	1	AIR CYLINDER	CLIPPERD	CL0002
2	2	E-CLIP	1/8"	EC0018
3	1	BACKPLATE	ALUMINUM	LGR001
4	1	1/4" × 2" BOLT	2" HARD BOLT	BT0002
5	1	SIDE FRAME (LF)	ALUMINUM	LGR003
6	11	CAP SCREW	6-32UNF-2 × 1/4"	CS0012
7	1	SLIDE PIN	STEEL	LGR006
8	1	CENTER BLOCK	ALUMINUM	LGR005
9	1	45° BEVEL GEAR	MODIFIED	BG0002
10	1	SPACER	ALUMINUM	LGR007
11	1	SIDE FRAME (RT)	ALUMINUM	LGR002
12	1	WASHER	BRASS	WA0014
13	1	PIVOT PIN	HARD STEEL	LGR008
14	1	45° BEVEL GEAR	BROWNING	BG0001

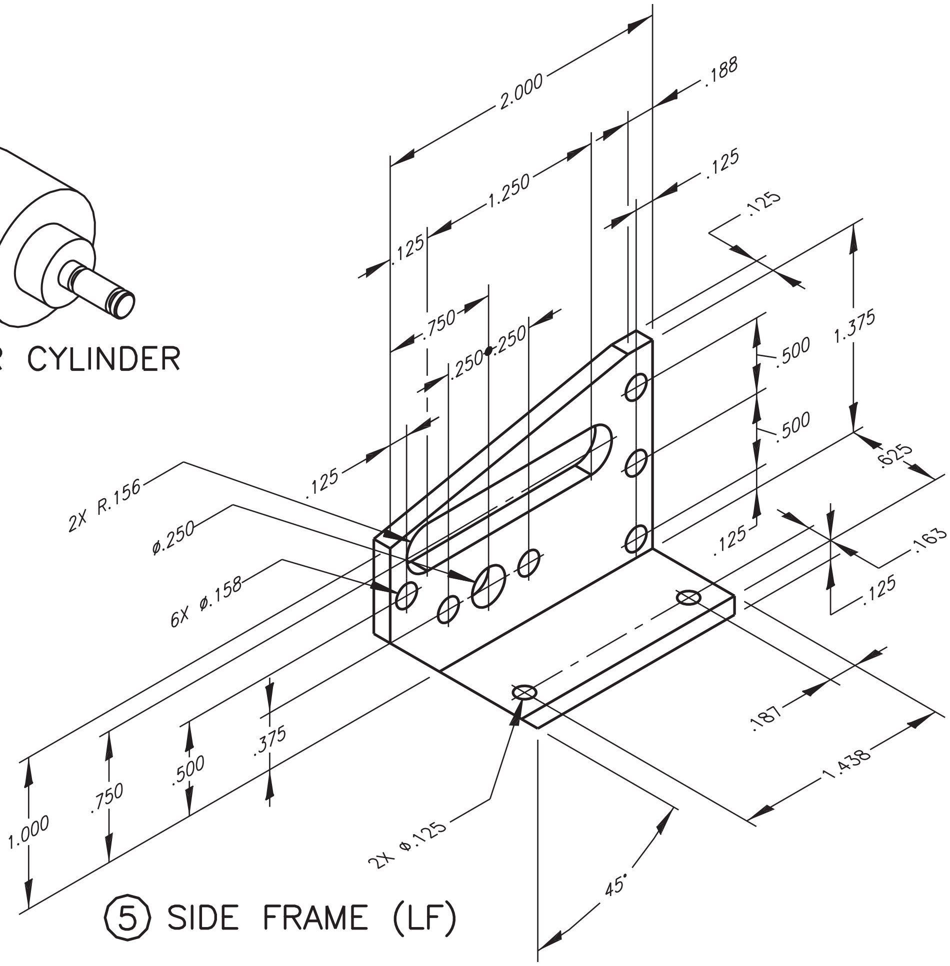


PROBLEM 15.19 *(Continued)*

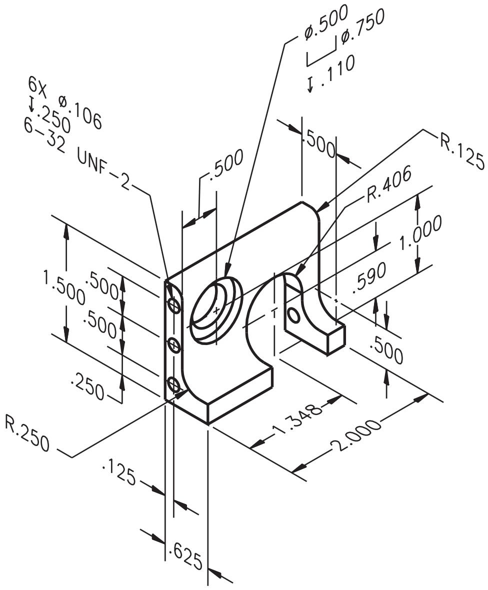


① AIR CYLINDER

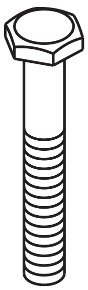
② E–CLIP



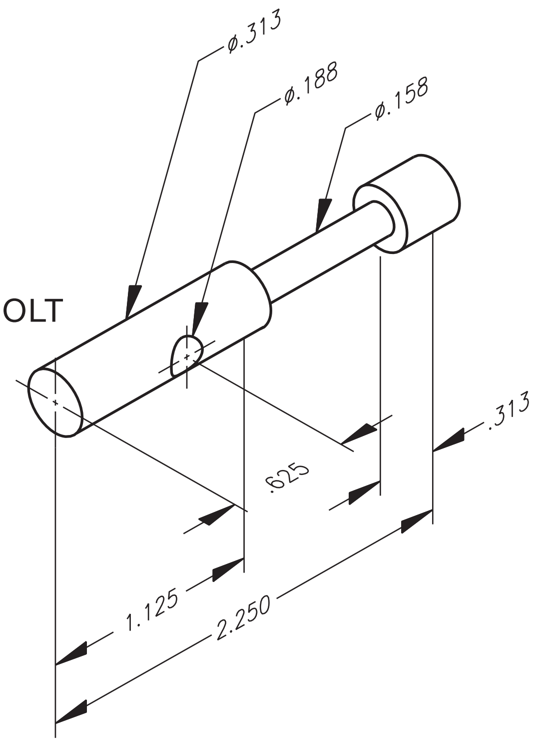
⑤ SIDE FRAME (LF)

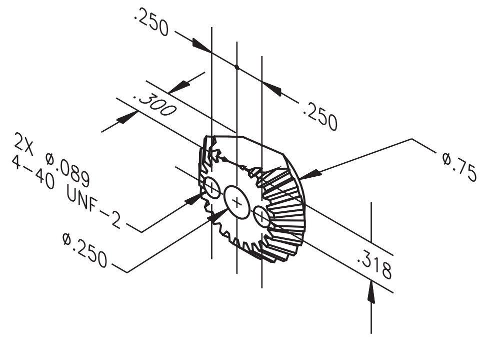


③ BACKPLATE

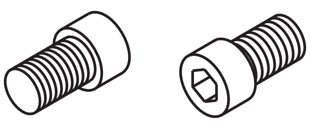


④ 1/4" X 2" BOLT





BEVEL GEAR DATA	
NUMBER OF TEETH	20
DIAMETRAL PITCH	20
PRESSURE ANGLE	20d
CONE DISTANCE	.707
PITCH DIAMETER	1.000
CIRCULAR THICKNESS (REF)	.07854
PITCH ANGLE	45d
ROOT ANGLE	40d
ADDENDEM	.05
WHOLE DEPTH	2.188
CHORDAL ADDENDEM	.052
CHORDAL THICKNESS	.078
DEDENDUM	.0625
OUTSIDE DIAMETER	1.071

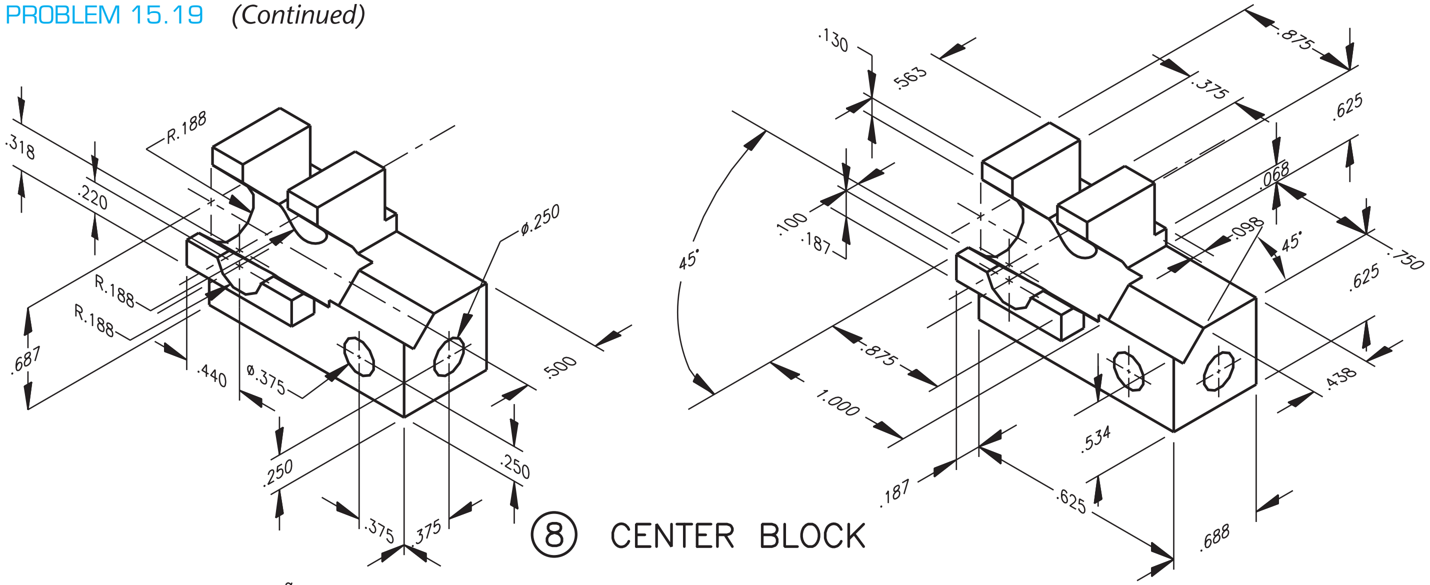


⑥ CAP SCREW
6–32 UNF–2 X 1/4"

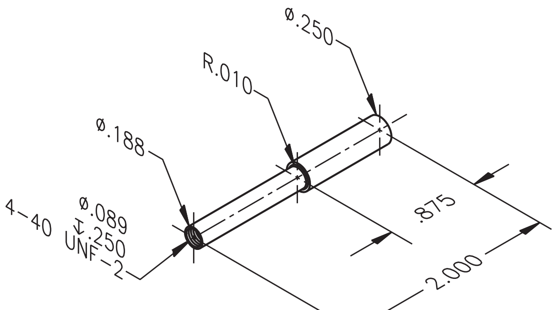
⑦ SLIDE PIN

⑨ 45° BEVEL GEAR (MODIFIED)

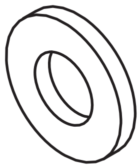
PROBLEM 15.19 *(Continued)*



(8) CENTER BLOCK

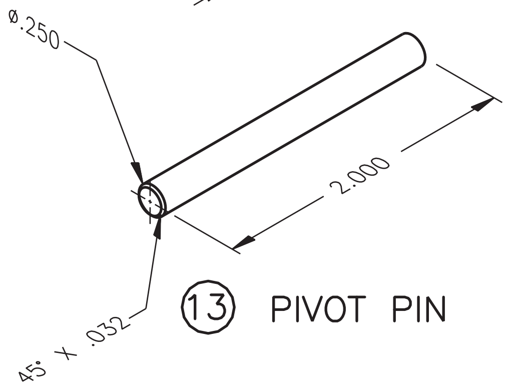


(10) SPACER

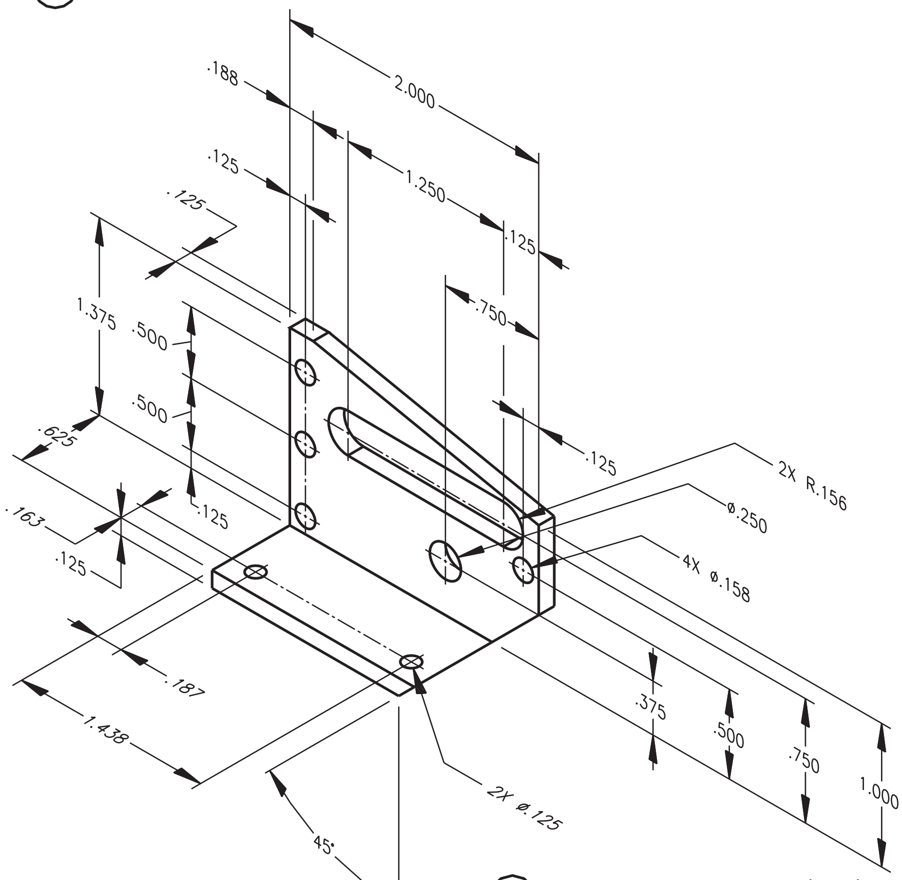


(12) WASHER

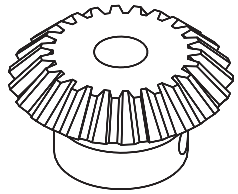
1/4" BRASS



(13) PIVOT PIN



(11) SIDE FRAME (RT)



BEVEL GEAR DATA	
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ADDENDEM	.05
WHOLE DEPTH	2.188
CHORDAL ADDENDEM	.052
CHORDAL THICKNESS	.078
DEDENDUM	.0625
OUTSIDE DIAMETER	1.071

(14) 45° BEVEL GEAR